INTRODUCTION

This document's purpose is to help the Project Manager and Technical Advisor (TA) through the planning stages of a Dry Low NOx (DLN) Gas Turbine installation. These individuals need to review and be cognizant of the following 10 base categories applicable to the unit.

The TA responsible for installing the DLN combustion system and related piping systems needs to identify responsible parties or individuals to help review and resolve any questions, concerns, discrepancies, or errors discovered during the planning process.

REFERENCE DRAWINGS AND BILL OF MATERIAL

1. Model List Items, Drawings and Schematics

ML 0306 - Mech. Outline, Gas Turbine and Load
ML 0313 - Customer Pipe Connection Drawings
ML 0323 - Foundation Interface, Pipe Supports
ML 0416 - Diagram, Schem. PP - Lube Oil †
ML 0417 - Diagram, Schem. PP - Clg. and Slg. Air
ML 0418 - Diagram, Schem. PP - Trip Oil †
ML 0419 - Diagram, Schem. PP - Control Air
ML 0420 - Diagram, Schem. PP - Cooling Water
ML 0422 - Diagram, Schem. PP - Fuel Gas
ML 0424 - Diagram, Schem. PP - Liquid Fuel **
ML 0425 - Diagram, Schem. PP – Atomizing Air
ML 0432 - Diagram, Schem. PP - Inlet Air Htg ** †
ML 0434 - Diagram, Schem. PP - Hydr. Supply †
ML 0462 - Diagram, Schem. PP - Water Inj. Sys **
ML 0477 - Diagram, Schem. PP - Fuel Purge
ML 0512 - Fuel Nozzle Assembly
ML 0513 - Gas Fuel Nozzle **
ML 0566 - Fuel Nozzle Assembly, Secondary
ML 0701 - Chamber Arrangement, Combustion
ML 0702 - Transition Piece Assembly Comb.
ML 0703 - Cap and Liner Arr., Comb.
ML 0717 - Transition Piece Arr., Comb.
ML 0719 - Case Arr., Comb. - Outer
ML 0722 - Cap, Combustion
ML 0909 - Piping Arr., Clg. and Slg. Air
ML 0915 - Piping Arr., Cooling Water Turbine
ML 0918 - Piping Arr., Fuel Oil Purge **
ML 0920 - Piping Arr., False Start Drain
ML 0924 - Piping Arr., Air Extraction
ML 0941 - Piping Arr., Steam Inj **
ML 0961 - Piping Arr., Fuel Oil Turbine **
ML 0962 - Piping Arr., Fuel Gas
ML 0963 - Piping Arr., Cooling Water (Turbine)
ML 0965 - Piping Arr., Atomizing Air
ML 0968 - Piping Arr., Water Injection **
ML 1121 - Flame Detector
ML 1214 - Spark Plug

** Does not apply to all units. Check ML list to verify if system is part of requisition.

† Does not apply to DLN system, though is recommended to have document accessible.

2. Complimentary Documents ††

P8A-AG1	Welding, General Specification
P8A-AG3	Welded Fabrication of Gas Turbine Piping
P24F-AG3	Air Flow Test, Gas Fuel Nozzle Assembly
P24E-AG2	Pressure Testing, Hydrostatic Fuel Nozzle
118D5700	Applied Practices
184A8937	Test Instructions, Spark Plugs
214A4435	Instruction, Assembly - Tube and Fittings
248A4158	Bolt and Stud Torquing (Elongation Table)
261A1671	Hydrostatic Test, Fuel Gas Piping
277A2415	Symbols, Piping
287A1300	Applied Practice, Conduit
307A9466	Instruction, Welding Visual Acceptance
314A5327	Instruction, Assembly - Automatic Bolt-Weld Fittings
349A6981	Instruction, Conduit Sealing
348A9200	Applied Practices, General Machining
351A3324	Instructions, Test - Fuel System
678D0872	Applied Practices - Piping

†† All Complimentary Drawings are GE PROPRIETARY INFORMATION and may not be used or disclosed to others except with written permission of General Electric.

3. Isometric Drawings

These drawings can be obtained through the Project Manager/Installation Engineer or GE Gas Turbines – Greenville. Obtain main piping systems, and the unit in its entirety (with or without piping).

4. Site Layout Drawings

Obtain General Arrangement and Plot Plan via Project Manager/Installation Engineer.

5. Installation and Shipping Schedule

Review the installation schedule and, if necessary, generate a bar chart for the DLN portion of the installation. Pay careful attention to installation sequence and installation of systems in parallel.

Review and compare the installation schedule with the availability and delivery of material and resources to ensure compatibility.

Listed below are existing I&FS planning guide support documents.

EM 5363 - MS7001FA Field Assembly and Installation of Gas Turbine DLN2 Combustion System and Piping.

EM 5535 - MS7001EA Field Assembly and Installation of Gas Turbine DLN1 Combustion System and Piping.

6. Review of Open Items

Obtain from and review with the Project Manager/Installation Engineer all MDRs, ANs, ECNs, and FMIs applicable to the unit that could impact the installation schedule, sequence, start-up and running of the unit.

7. Model List

The unit Model List has a PIN identification nomenclature assigned to each ML item. For example, DLN combustion components ML 0701, ML 0703, and ML 0719 may have a PIN reference of TE01. TE01 indicates that the combustion components are required for testing in Greenville and do not ship with the unit or apply to field installation. The following are other commonly used PIN numbers.

- TE01 - Unit assembly, Combustion Test
- PT01 - Combustion prep to ship

- UB34 - Turbine Base, ready for test
- UB50 - Base assembly, Turbine Lower
- DLND - Dry Low NOx Summary Containers
- MSD - Material Shipped Direct
- SUMM - Summary Sheet Container

8. DLN Standard Summary Containers

Material that fails to ship on the unit or as MSD will ship in one of two ways; Standard Summary Container, or Dry Low NOx Summary Containers. To verify that all material is accounted for, the TA should compare the packing list for these containers to the Bill of Material for the ML items listed on the unit Model List.

9. DLN Tuning

Some units require mechanical DLN Tuning Valves. The TA should review and answer the following questions regarding this issue.

1. Does the unit require Tuning Valves?
2. Are the valves factory installed or require field installation?
3. Who installs and tuning the valves and what process is required?

10. Flow Testing

The Project Manager and TA should be cognizant that some units and combustion components have required flow testing in the past. Review and determine when, where and who performed this testing if applicable to the unit.